Problem Set 3 Solutions

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Problem 1: Directed Acyclic Garlic Bread

A pizza oven has an unknown calibration described by two learnable parameters a and b. For pizza i, the true (unobserved) oven temperature is

$$T_i \sim \text{Norm}\big(\mu = aT_{\text{set}}^{(i)} + b,\ \sigma^2 = 5^2\big),$$

which is constant throughout the cooking process for a particular pizza.

Define the "pizza cooking completion score"

$$H_i = \frac{T_i}{200} \cdot \frac{t_i}{15} \cdot \left(\frac{2}{d_i}\right)^2,$$

where t_i is bake time in minutes and d_i is dough thickness in cm.

We say:

$$\begin{cases} \text{undercooked} & \text{if } H_i < 0.9 \\ \text{perfect} & \text{if } 0.9 \leq H_i \leq 1.1 \\ \text{overcooked} & \text{if } H_i > 1.1 \end{cases}$$

The observed dataset is:

pizza	T_{set}	t	d	outcome
1	190	15.5	2.0	perfect
2	205	15.0	2.2	under
3	215	14.0	1.9	perfect
4	220	16.0	2.0	over
5	230	16.0	1.9	over

Let the observed class label be

$$y_i \in \{\text{under}, \text{perfect}, \text{over}\}.$$

(a) Draw the directed PGM, using a plate over pizzas.

The unknown calibration parameters a and b determine the true oven temperature T_i.

The set temperature $T_{\text{set}}^{(i)}$ also influences T_i.

The class label y_i depends on:

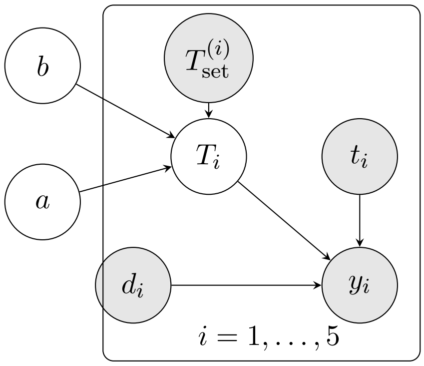
- the true temperature T_i
- bake time t_i
- dough thickness d_i

Thus the dependencies are

$$a, b, T_{\text{set}}^{(i)} \to T_i$$

and

$$T_i, t_i, d_i \to y_i$$



Observed vs. latent variables in the PGM. In PGMs we typically visually distinguish between *observed variables* and *latent (hidden) variables*. Observed variables correspond to quantities that are directly measured in the dataset, while latent variables represent unknown quantities that are not observed but are inferred by the model (a, d, T_i).

The parameters a and b represent the unknown calibration of the oven and must be learned from the data. The variable T_i represents the true oven temperature for pizza i, which is not directly observed but is modeled probabilistically.

(b) Treat a and b as random variables with unspecified priors, and treat $T_{\text{set}}^{(i)}, t_i, d_i$ as observed root-node variables whose factors may be left symbolic.

Write the full joint

$$P(a, b, T_{1:5}, y_{1:5}, T_{\text{set},1:5}, t_{1:5}, d_{1:5})$$

in factorized form.

Factorization of the joint distribution

A fundamental property of PGMs is that the joint distribution of all variables factorizes according to the parent-child relationships in the graph.

So, if $X_1, \ldots, X_n$ are the variables in a directed graphical model, the joint distribution factorizes as

$$P(X_1, \ldots, X_n) = \prod_{k=1}^{n} P(X_k \mid \text{Parents}(X_k)).$$

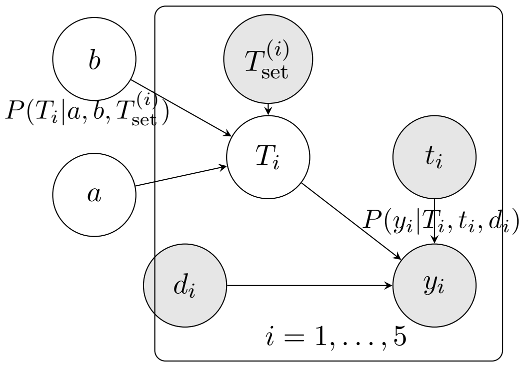
Therefore, each variable contributes one factor consisting of its conditional distribution given its parents in the graph.

Now, let's do that for our case: we need to take the PGM from section a), and for every node X in the graph, we include a factor

$$P(X \mid \text{Parents}(X)).$$

and then we can read the factorization directly from the arrows in the graph.

Graphical model with conditional probabilities



From this diagram we can identify the parents of each variable.

- The global parameters a and b have no parents in the graph, so their contributions are simply
$$P(a), \qquad P(b).$$
- The observed quantities $T_{\text{set}}^{(i)}$, t_i, and d_i are root nodes of the graph. The problem statement tells us that their distributions can be left symbolic, so we write
$$P(T_{\text{set}}^{(i)}), \quad P(t_i), \quad P(d_i).$$
- The latent temperature T_i depends on its parents a, b, and $T_{\text{set}}^{(i)}$. Therefore its contribution to the joint distribution is
$$P(T_i \mid T_{\text{set}}^{(i)}, a, b).$$
- Finally, the observed outcome y_i depends on T_i, t_i, and d_i, so we include the factor
$$P(y_i \mid T_i, t_i, d_i).$$

Since the pizzas are conditionally independent given the parameters, we multiply the contributions for each pizza i (this is represented by the plate in the graph, the same graphical structure applies to each pizza independently).

Putting all factors together gives the full joint distribution

$$P(a, b, T_{1:5}, y_{1:5}, T_{\text{set},1:5}, t_{1:5}, d_{1:5})$$

$$= P(a)P(b)\prod_{i=1}^{5} P(T_{\text{set}}^{(i)})P(t_i)P(d_i)P(T_i|T_{\text{set}}^{(i)}, a, b)P(y_i|T_i, t_i, d_i).$$

Note that the parameters a and b appear only once in the joint distribution. This is because they are global parameters shared by all pizzas and therefore exist only once in the model, it can be seen because they are outside the rectangle, and that they appear as a and b instead of a_i and b_i, this is the power of PGMs, you write it down once, which is quite intuitive, adn then when you factorize the fully joint probability distribution, it's easier to do it correctly and don't count things the wrong amount of times.

Let's look at the form of the two conditional probabilities that appear in our full joint distirbution For the distribution of the latent temperature, the problem statement specifies that the true temperature follows

$$T_i \sim \mathcal{N}\big(aT_{\text{set}}^{(i)} + b,\ 5^2\big).$$

with

$$\mu_i = aT_{\text{set}}^{(i)} + b$$

and

$$\sigma^2 = 5^2$$

Substituting the mean and variance of our model gives

$$P(T_i|T_{\text{set}}^{(i)}, a, b) = \mathcal{N}\big(T_i; \mu_i =\ aT_{\text{set}}^{(i)} + b, \sigma^2 = 5^2\big).$$

The second conditional probability appearing in the joint distribution, the distribution of the observed outcome, is

$$P(y_i \mid T_i, t_i, d_i).$$

Unlike the temperature distribution, this probability is not directly given as a standard probability distribution in the problem statement. Instead, it is defined through the pizza completion score

$$H_i = \frac{T_i}{200} \cdot \frac{t_i}{15} \cdot \left(\frac{2}{d_i}\right)^2.$$

The observed outcome y_i is determined by comparing this score to two thresholds:

$$y_i = \begin{cases} \text{under} & H_i < 0.9 \\ \text{perfect} & 0.9 \le H_i \le 1.1 \\ \text{over} & H_i > 1.1 \end{cases}$$

This means that once T_i, t_i, and d_i are known, the value of H_i is completely determined. Therefore the label y_i is also completely determined.

In probabilistic terms, this means that y_i is a deterministic function of T_i, t_i, d_i.

We can therefore write

$$P(y_i \mid T_i, t_i, d_i) = \begin{cases} 1 & \text{if the value of } y_i \text{ matches the rule above} \\ 0 & \text{otherwise.} \end{cases}$$

So the conditional probability simply enforces the rule that maps the continuous score H_i into one of the three possible categories.

(c) By marginalizing over the latent temperature T_i, write the negative log-likelihood

$$L[a, b] = -\sum_{i=1}^{5} \log P(y_i | T_{\text{set}}^{(i)}, t_i, d_i, a, b).$$

Likelihood of the observed data

The dataset consists of the observed variables

$$(T_{\text{set}}^{(i)}, t_i, d_i, y_i)$$

for each pizza i, but the latent temperature T_i is not observed.

Therefore, when estimating the parameters a and b, we must compute the probability of the observed label y_i conditioned on the observed inputs and the model parameters.

The likelihood contribution of a single data point is therefore

$$P(y_i \mid T_{\text{set}}^{(i)}, t_i, d_i, a, b).$$

Since pizzas are conditionally independent given the parameters, the likelihood of the full dataset becomes

$$\prod_{i=1}^{5} P(y_i \mid T_{\text{set}}^{(i)}, t_i, d_i, a, b).$$

Taking the negative logarithm gives the negative log-likelihood

$$L[a, b] = -\sum_{i=1}^{5} \log P(y_i \mid T_{\text{set}}^{(i)}, t_i, d_i, a, b).$$

Therefore we need to compute the probability

$$P(y_i \mid T_{\text{set}}^{(i)}, t_i, d_i, a, b),$$

which requires marginalizing over the latent temperature T_i.

Marginalizing the latent temperature

Recall the general rule of marginalization: if X denotes a collection of variables and Z is a latent variable,

$$P(Y|X) = \int P(Y, Z|X)\, dZ.$$

This is the continuous version of the law of total probability.

In our case we define

$$X = (T_{\text{set}}^{(i)}, t_i, d_i, a, b),$$

and the latent variable is T_i. Therefore

$$P(y_i|T_{\text{set}}^{(i)}, t_i, d_i, a, b) = \int P(y_i, T_i|T_{\text{set}}^{(i)}, t_i, d_i, a, b)\, dT_i.$$

Using the chain rule of probability,

$$P(Y, Z|X) = P(Y|Z, X)P(Z|X),$$

we obtain

$$P(y_i|T_{\text{set}}^{(i)}, t_i, d_i, a, b) = \int P(y_i|T_i, T_{\text{set}}^{(i)}, t_i, d_i, a, b)P(T_i|T_{\text{set}}^{(i)}, t_i, d_i, a, b)\, dT_i.$$

Using conditional independences from the PGM

From the graphical model we know that

- y_i depends only on T_i, t_i, d_i
- T_i depends only on $T_{\text{set}}^{(i)}, a, b$

Therefore

$$P(y_i|T_i, T_{\text{set}}^{(i)}, t_i, d_i, a, b) = P(y_i|T_i, t_i, d_i)$$

and

$$P(T_i|T_{\text{set}}^{(i)}, t_i, d_i, a, b) = P(T_i|T_{\text{set}}^{(i)}, a, b).$$

Substituting these simplifications gives

$$P(y_i|T_{\text{set}}^{(i)}, t_i, d_i, a, b) = \int P(y_i|T_i, t_i, d_i)\, P(T_i|T_{\text{set}}^{(i)}, a, b)\, dT_i.$$

The second term,

$$P(T_i|T_{\text{set}}^{(i)}, a, b),$$

is the Gaussian density for the latent temperature T_i. This tells us how likely each possible value of T_i is under the model.

The first term,

$$P(y_i|T_i, t_i, d_i),$$

, as we saw in b), is determined by the deterministic rule that maps the score H_i to the class label y_i . Once T_i, t_i, and d_i are fixed, the score H_i is fixed, and therefore the label is fixed as well. So this term is equal to 1 when the chosen value of T_i is consistent with the observed class, and 0 otherwise. So it acts like an indicator function.

Expressing the class boundaries in terms of T_i

Recall the pizza completion score

$$H_i = c_i T_i$$

with

$$c_i = \frac{1}{200} \cdot \frac{t_i}{15} \cdot \left(\frac{2}{d_i}\right)^2.$$

The classification rules are

$$\text{under if } H_i < 0.9, \qquad \text{perfect if } 0.9 \le H_i \le 1.1, \qquad \text{over if } H_i > 1.1.$$

Substituting $H_i = c_i T_i$ gives thresholds on T_i:

$$T_i < \frac{0.9}{c_i}, \quad \frac{0.9}{c_i} \le T_i \le \frac{1.1}{c_i}, \quad T_i > \frac{1.1}{c_i}.$$

Define

$$\tau_i^- = \frac{0.9}{c_i}, \qquad \tau_i^+ = \frac{1.1}{c_i}.$$

So, for example, if the observed label is $y_i = \text{under}$, then

$$P(y_i = \text{under}|T_i, t_i, d_i) = \begin{cases} 1 & \text{if } T_i < \tau_i^- \\ 0 & \text{otherwise.} \end{cases}$$

Substituting this into the integral gives

$$P(y_i = \text{under}|T_{\text{set}}^{(i)}, t_i, d_i, a, b) = \int_{-\infty}^{\tau_i^-} P(T_i|T_{\text{set}}^{(i)}, a, b)\, dT_i.$$

So the indicator function has simply restricted the region of integration: instead of integrating over all values of T_i, we only integrate over the values that are compatible with the observed class.

Similarly, if the observed label is perfect, then we integrate only over the interval where $\tau_i^- \leq T_i \leq \tau_i^+$, and if the observed label is over, then we integrate only over the region $T_i > \tau_i^+$.

Using the Gaussian distribution of T_i

From the model

$$T_i \sim \mathcal{N}(\mu_i, 5^2), \qquad \mu_i = aT_{\text{set}}^{(i)} + b.$$

Therefore the integrals above are Gaussian probabilities over intervals. These are exactly the kinds of probabilities computed by the cumulative distribution function (CDF).

Recall that if

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

then

$$P(X \leq t) = \int_{-\infty}^{t} \mathcal{N}(x; \mu, \sigma^2)\, dx.$$

This is precisely the Gaussian CDF evaluated at t. To compute it in a standard form, we standardize the Gaussian by introducing

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim \mathcal{N}(0, 1).$$

Hence

$$P(X \leq t) = P\left(Z \leq \frac{t - \mu}{\sigma}\right) = \Phi\left(\frac{t - \mu}{\sigma}\right),$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution.

Applying this to our model, with mean μ_i and standard deviation 5, gives

$$P(y_i = \text{under}) = P(T_i < \tau_i^-) = \Phi\left(\frac{\tau_i^- - \mu_i}{5}\right).$$

For the perfect class, we need the probability that T_i lies between two thresholds, so we subtract two CDF values:

$$P(y_i = \text{perfect}) = P(\tau_i^- \leq T_i \leq \tau_i^+) = \Phi\left(\frac{\tau_i^+ - \mu_i}{5}\right) - \Phi\left(\frac{\tau_i^- - \mu_i}{5}\right).$$

For the overcooked class, we need the right tail of the Gaussian, so we take the complement of the left-tail probability:

$$P(y_i = \text{over}) = P(T_i > \tau_i^+) = 1 - \Phi\left(\frac{\tau_i^+ - \mu_i}{5}\right).$$

Negative log-likelihood

Let $p_i[a, b]$ denote the probability corresponding to the observed label y_i. The negative log-likelihood becomes

$$L[a, b] = -\sum_{i=1}^{5} \log p_i[a, b].$$

Equivalently

$$L[a, b] = -\sum_{i:y_i=\text{under}} \log \Phi\left(\frac{\tau_i^- - \mu_i}{5}\right)$$

$$-\sum_{i:y_i=\text{perfect}} \log\left[\Phi\left(\frac{\tau_i^+ - \mu_i}{5}\right) - \Phi\left(\frac{\tau_i^- - \mu_i}{5}\right)\right]$$

$$-\sum_{i:y_i=\text{over}} \log\left[1 - \Phi\left(\frac{\tau_i^+ - \mu_i}{5}\right)\right].$$

(d) Run an optimizer on a and b and report the maximum-likelihood estimate. Using what you have found, what temperature should you set the oven to so that the expected true oven temperature is $T = 200$?

Optimizing the negative log-likelihood

From part (c), we derived the negative log-likelihood

$$L[a, b] = -\sum_{i=1}^{5} \log p_i[a, b],$$

where $p_i[a, b]$ is the probability assigned by the model to the observed label of pizza i.

We obtain the MLE numerically using an optimizer, since the expression contains Gaussian CDFs and does not simplify to a closed-form solution.

Running the optimizer gives

$$\boxed{\hat{a} \approx 0.916, \qquad \hat{b} \approx 10.64.}$$

So the fitted model for the latent temperature is

$$T_i \sim \mathcal{N}\big(\hat{a}\, T_{\text{set}}^{(i)} + \hat{b},\ 5^2\big) = \mathcal{N}\big(0.916\, T_{\text{set}}^{(i)} + 10.64,\ 5^2\big).$$

Interpreting the fitted parameters

Recall that in our model the true oven temperature has mean

$$\mu_i = aT_{\text{set}}^{(i)} + b.$$

So after fitting, the expected latent oven temperature corresponding to a given set temperature T_{set} is

$$\mathbb{E}[T_i \mid T_{\text{set}}] = \hat{a}\, T_{\text{set}} + \hat{b}.$$

This tells us how the oven setting translates into the actual average temperature inside the oven after accounting for calibration error.

Choosing the oven setting so that the expected true temperature is 200°C

We want the expected latent oven temperature to be exactly 200°C. That means we want the Gaussian mean to satisfy

$$\hat{a}\, T_{\text{set}} + \hat{b} = 200.$$

Now solve for the required set temperature:

$$\boxed{T_{\text{set}}^{*} = \frac{200 - \hat{b}}{\hat{a}} = \frac{200 - 10.64}{0.916} \approx 206.8^{\circ}\text{C}.}$$

Problem 2. ELBO Any% Speedrun

Show that the ELBO can be derived from the KL divergence between the variational posterior and the true posterior:

$$D_{\text{KL}}[q(z|x,\theta) \,\|\, \Pr(z|x,\phi)] = \int q(z|x,\theta) \log\left(\frac{q(z|x,\theta)}{\Pr(z|x,\phi)}\right) dz.$$

Specifically, use Bayes' rule to show that

$$\text{ELBO}[\theta,\phi] = \int q(z|x,\theta) \log\left(\frac{\Pr(x,z|\phi)}{q(z|x,\theta)}\right) dz$$

$$= \mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)] - D_{\text{KL}}[q(z|x,\theta) \,\|\, \Pr(z)]$$

is a lower bound on $\log \Pr(x|\phi)$.

We begin from the Kullback–Leibler (KL) divergence between the variational posterior $q(z|x,\theta)$ and the true posterior $\Pr(z|x,\phi)$.

The KL divergence is a measure of how different two probability distributions are. In general, for two distributions $q(z)$ and $p(z)$, it is defined as

$$D_{\mathrm{KL}}[q(z) \,\|\, p(z)] = \int q(z) \log\left(\frac{q(z)}{p(z)}\right) dz.$$

It can be interpreted as the expected log-difference between the two distributions when the expectation is taken with respect to $q(z)$. It's always nonnegative,

$$D_{\mathrm{KL}}[q(z) \,\|\, p(z)] \geq 0,$$

and it is equal to zero only when the two distributions are identical almost everywhere.

For variational inference problems, the true posterior is typically intractable to compute exactly, so we introduce a simpler distribution $q(z|x,\theta)$ and try to make it as close as possible to the true posterior by minimizing this KL divergence.

Therefore we consider

$$D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z|x,\phi)] = \int q(z|x,\theta) \log\left(\frac{q(z|x,\theta)}{\Pr(z|x,\phi)}\right) dz.$$

And now we are asked to manipulate this expression using Bayes' rule so that the Evidence Lower Bound (ELBO) appears explicitly (used for VI implementations).

Using Bayes' rule,

$$\Pr(z|x,\phi) = \frac{\Pr(x|z,\phi)\Pr(z)}{\Pr(x|\phi)}.$$

where:

- the likelihood $\Pr(x|z,\phi)$,
- the prior $\Pr(z)$,
- and the marginal likelihood $\Pr(x|\phi)$.

Substituting this into the KL divergence gives

$$D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z|x,\phi)] = \int q(z|x,\theta) \log\left(\frac{q(z|x,\theta)}{\frac{\Pr(x|z,\phi)\Pr(z)}{\Pr(x|\phi)}}\right) dz = \int q(z|x,\theta) \log\left(\frac{\Pr(x|\phi)\, q(z|x,\theta)}{\Pr(x|z,\phi)\Pr(z)}\right) dz.$$

Now using:

$$\log\left(\frac{AB}{CD}\right) = \log A + \log B - \log C - \log D.$$

we get:

$$D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z|x,\phi)] = \int q(z|x,\theta) \left[\log \Pr(x|\phi) + \log q(z|x,\theta) - \log \Pr(x|z,\phi) - \log \Pr(z)\right] dz.$$

so:

$$D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z|x,\phi)] = \int q(z|x,\theta) \log \Pr(x|\phi) \, dz$$

$$+ \int q(z|x,\theta) \log q(z|x,\theta) \, dz - \int q(z|x,\theta) \log \Pr(x|z,\phi) \, dz - \int q(z|x,\theta) \log \Pr(z) \, dz.$$

The quantity $\Pr(x|\phi)$ does not depend on z, so it is constant with respect to the integration variable. Therefore it can be pulled out of the integral:

$$\int q(z|x,\theta) \log \Pr(x|\phi) \, dz = \log \Pr(x|\phi) \int q(z|x,\theta) \, dz.$$

Since $q(z|x,\theta)$ is a probability distribution, it integrates to 1. Hence

$$\int q(z|x,\theta) \log \Pr(x|\phi) \, dz = \log \Pr(x|\phi).$$

So we now have

$$D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z|x,\phi)] = \log \Pr(x|\phi) + \int q(z|x,\theta) \log q(z|x,\theta) \, dz$$

$$- \int q(z|x,\theta) \log \Pr(x|z,\phi) \, dz - \int q(z|x,\theta) \log \Pr(z) \, dz.$$

Recognize the KL term and the expectation

Now group the second and fourth terms:

$$\int q(z|x,\theta) \log q(z|x,\theta) \, dz - \int q(z|x,\theta) \log \Pr(z) \, dz$$

$$= \int q(z|x,\theta) \log\left(\frac{q(z|x,\theta)}{\Pr(z)}\right) dz = D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z)].$$

And the third term is exactly an expectation:

$$\int q(z|x,\theta) \log \Pr(x|z,\phi) \, dz = \mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)].$$

Therefore

$$D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z|x,\phi)] = \log \Pr(x|\phi) + D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z)] - \mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)].$$

Rearranging the equation to isolate the ELBO gives

$$\mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)] - D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z)] = \log \Pr(x|\phi) - D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z|x,\phi)].$$

Identifying the ELBO

The LHS term is called the *Evidence Lower Bound* (ELBO) and is defined as

$$\mathrm{ELBO}[\theta,\phi] = \mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)] - D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z)].$$

The first term measures how well the latent variable z explains the observed data x under the model (often called the *reconstruction term*), while the second term penalizes the divergence between the variational distribution $q(z|x,\theta)$ and the prior $\Pr(z)$.

Writing the ELBO as a single integral

We can rewrite the ELBO in a more compact form by expanding both terms.

An expectation with respect to q can be written as an integral:

$$\mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)] = \int q(z|x,\theta) \log \Pr(x|z,\phi)\, dz.$$

The KL divergence can also be written in integral form:

$$D_{\mathrm{KL}}[q(z|x,\theta) \parallel \Pr(z)] = \int q(z|x,\theta) \log \frac{q(z|x,\theta)}{\Pr(z)}\, dz.$$

Substituting these expressions into the ELBO gives

$$\mathrm{ELBO} = \int q(z|x,\theta) \left[\log \Pr(x|z,\phi) - \log \frac{q(z|x,\theta)}{\Pr(z)}\right] dz.$$

Using logarithm identities we combine the terms inside the brackets:

$$\log \Pr(x|z,\phi) + \log \Pr(z) - \log q(z|x,\theta).$$

Now recall the definition of the joint distribution:

$$\Pr(x,z|\phi) = \Pr(x|z,\phi) \Pr(z).$$

Using this identity we obtain the equivalent form of the ELBO

$$\mathrm{ELBO}[\theta,\phi] = \int q(z|x,\theta) \log\left(\frac{\Pr(x,z|\phi)}{q(z|x,\theta)}\right) dz.$$

Now we show that this is a lower bound

From the definition of KL divergence, we always have

$$D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z|x,\phi)] \geq 0.$$

Using the rearranged formula

$$\mathrm{ELBO}[\theta,\phi] = \log \Pr(x|\phi) - D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z|x,\phi)],$$

we immediately obtain

$$\mathrm{ELBO}[\theta,\phi] \leq \log \Pr(x|\phi).$$

Therefore the ELBO is indeed a lower bound on the log marginal likelihood (the probability that our model assigns to the data point x, which is what we want to maximize, but since $\Pr(x|\phi)$is high-dimensional and untractable, therefore we maximize the ELBO, and since it's a lower bound for $\Pr(x|\phi)$, we are also maximizing the marginal log likelihood.

Final result

$$\boxed{\mathrm{ELBO}[\theta,\phi] = \mathbb{E}_{q(z|x,\theta)}[\log \Pr(x|z,\phi)] - D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z)]}$$

and equivalently

$$\boxed{\mathrm{ELBO}[\theta,\phi] = \log \Pr(x|\phi) - D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z|x,\phi)]}$$

so, since KL divergence is nonnegative,

$$\boxed{\mathrm{ELBO}[\theta,\phi] \leq \log \Pr(x|\phi)}.$$

Problem 3. Mind the Gap

Let

$$\Pr(z) = \mathcal{N}(z;\mu_p,\sigma_p^2), \qquad q(z) = \mathcal{N}(z;\mu_q,\sigma_q^2),$$

with $\sigma_p^2 > 0$ and $\sigma_q^2 > 0$. Prove that the KL divergence, defined by

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \int q(z) \log\left(\frac{q(z)}{\Pr(z)}\right) dz,$$

takes the form

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \log\left(\frac{\sigma_p}{\sigma_q}\right) + \frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{2\sigma_p^2} - \frac{1}{2}.$$

We want to compute the KL divergence between two one-dimensional Gaussian distributions,

$$q(z) = \mathcal{N}(z; \mu_q, \sigma_q^2), \qquad \Pr(z) = \mathcal{N}(z; \mu_p, \sigma_p^2).$$

And we have that:

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \int q(z) \log\left(\frac{q(z)}{\Pr(z)}\right) dz.$$

So we need to expand the Gaussian densities, simplify the logarithm, and then take the expectation under $q(z)$.

We have:

$$q(z) = \frac{1}{\sqrt{2\pi\sigma_q^2}} \exp\left(-\frac{(z-\mu_q)^2}{2\sigma_q^2}\right),$$

and

$$\Pr(z) = \frac{1}{\sqrt{2\pi\sigma_p^2}} \exp\left(-\frac{(z-\mu_p)^2}{2\sigma_p^2}\right).$$

Therefore:

$$\log\left(\frac{q(z)}{\Pr(z)}\right) = \log\left(\frac{\frac{1}{\sqrt{2\pi\sigma_q^2}} \exp\left(-\frac{(z-\mu_q)^2}{2\sigma_q^2}\right)}{\frac{1}{\sqrt{2\pi\sigma_p^2}} \exp\left(-\frac{(z-\mu_p)^2}{2\sigma_p^2}\right)}\right).$$

Now we simplify this expression term by term. The ratio of the normalization constants becomes

$$\frac{1/\sqrt{2\pi\sigma_q^2}}{1/\sqrt{2\pi\sigma_p^2}} = \frac{\sigma_p}{\sigma_q},$$

and when dividing exponentials, the exponents subtract. Therefore

$$\log\left(\frac{q(z)}{\Pr(z)}\right) = \log\left(\frac{\sigma_p}{\sigma_q}\right) - \frac{(z-\mu_q)^2}{2\sigma_q^2} + \frac{(z-\mu_p)^2}{2\sigma_p^2}.$$

By definition of KL divergence,

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \mathbb{E}_{q(z)}\left[\log\left(\frac{q(z)}{\Pr(z)}\right)\right].$$

Substituting the expression above,

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \mathbb{E}_{q(z)}\left[\log\left(\frac{\sigma_p}{\sigma_q}\right) - \frac{(z-\mu_q)^2}{2\sigma_q^2} + \frac{(z-\mu_p)^2}{2\sigma_p^2}\right].$$

Since $\log(\sigma_p/\sigma_q)$ is constant with respect to z, it comes out of the expectation:

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \log\left(\frac{\sigma_p}{\sigma_q}\right) - \frac{1}{2\sigma_q^2}\mathbb{E}_{q(z)}[(z-\mu_q)^2] + \frac{1}{2\sigma_p^2}\mathbb{E}_{q(z)}[(z-\mu_p)^2].$$

So now we just need to compute the two expectations.

Since $q(z)$ is a Gaussian with mean μ_q and variance σ_q^2,

$$\mathbb{E}_{q(z)}[(z-\mu_q)^2] = \sigma_q^2.$$

This is just the definition of the variance.

For this term: $\mathbb{E}_{q(z)}[(z-\mu_p)^2]$ it's not so direct, because the expectation is still taken with respect to $q(z)$, whose mean is μ_q, not μ_p.

We rewrite

$$z - \mu_p = (z-\mu_q) + (\mu_q - \mu_p).$$

Now square both sides:

$$(z-\mu_p)^2 = (z-\mu_q)^2 + 2(\mu_q-\mu_p)(z-\mu_q) + (\mu_q-\mu_p)^2.$$

Taking the expectation with respect to $q(z)$ gives

$$\mathbb{E}_{q(z)}[(z-\mu_p)^2] = \mathbb{E}_{q(z)}[(z-\mu_q)^2] + 2(\mu_q-\mu_p)\mathbb{E}_{q(z)}[z-\mu_q] + (\mu_q-\mu_p)^2.$$

Now use two standard facts for a Gaussian with mean μ_q:

$$\mathbb{E}_{q(z)}[(z-\mu_q)^2] = \sigma_q^2, \qquad \mathbb{E}_{q(z)}[z-\mu_q] = 0.$$

Therefore the middle term vanishes, and we obtain

$$\mathbb{E}_{q(z)}[(z-\mu_p)^2] = \sigma_q^2 + (\mu_q-\mu_p)^2.$$

Substituting the two expectation values into the expression for the KL divergence gives

$$D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \log\left(\frac{\sigma_p}{\sigma_q}\right) - \frac{1}{2\sigma_q^2}\sigma_q^2 + \frac{1}{2\sigma_p^2}\left[\sigma_q^2 + (\mu_q-\mu_p)^2\right].$$

So we get

$$\boxed{D_{\mathrm{KL}}[q(z) \,\|\, \Pr(z)] = \log\left(\frac{\sigma_p}{\sigma_q}\right) + \frac{\sigma_q^2 + (\mu_q-\mu_p)^2}{2\sigma_p^2} - \frac{1}{2}}$$

Interpretation

- the term
$$\log\left(\frac{\sigma_p}{\sigma_q}\right)$$
compares the spreads of the two Gaussians,
- the term
$$\frac{(\mu_q-\mu_p)^2}{2\sigma_p^2}$$
penalizes the difference in their means,
- the term
$$\frac{\sigma_q^2}{2\sigma_p^2}$$
penalizes the mismatch in their variances.

So the KL divergence becomes large when the two Gaussians have very different centers or very different spreads.

Problem 4. (I can't do math)

For one image x, a VAE encoder returns

$$\mu[x,\theta]=\begin{bmatrix}1\\-0.5\end{bmatrix},\qquad \log\sigma^2[x,\theta]=\begin{bmatrix}-1\\ \log 4\end{bmatrix}.$$

Assume the approximate posterior is diagonal Gaussian and the prior is standard normal:

$$q(z|x,\theta)=\mathcal{N}\big(z;\,\mu[x,\theta],\mathrm{Diag}[\sigma_1^2,\sigma_2^2]\big)\,,\qquad \Pr(z)=\mathcal{N}(z;0,I).$$

Some theory:

KL divergence for a diagonal Gaussian posterior (VAE case). In variational autoencoders (VAEs) we approximate the intractable posterior distribution $p(z|x)$ with a simpler distribution $q(z|x,\theta)$ parameterized by a neural network. Training the model corresponds to maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}(x)=\mathbb{E}_{q(z|x,\theta)}[\log p(x|z)]-D_{\mathrm{KL}}(q(z|x,\theta)\,\|\,p(z)).$$

The first term encourages accurate reconstruction of the data, while the KL divergence term acts as a regularizer that encourages the approximate posterior $q(z|x,\theta)$ to stay close to the prior $p(z)$.

In the standard VAE setting we assume that the approximate posterior is a Gaussian with diagonal covariance,

$$q(z|x,\theta)=\mathcal{N}(\mu(x),\,\mathrm{diag}(\sigma^2(x))),$$

and that the latent prior is a standard normal distribution,

$$p(z) = \mathcal{N}(0, I).$$

Because both distributions are Gaussian, the KL divergence can be computed analytically, avoiding the need for Monte Carlo estimation.

KL divergence definition for two distributions $q(z)$ and $p(z)$:

$$D_{\mathrm{KL}}(q\|p) = \mathbb{E}_{q(z)}\left[\log \frac{q(z)}{p(z)}\right].$$

It measures how much information is lost when $p(z)$ is used to approximate $q(z)$. Reminder: the KL divergence is always non-negative and is zero only if the two distributions are identical.

General KL formula for Gaussians. If both distributions are multivariate Gaussians,

$$q(z) = \mathcal{N}(\mu_q, \Sigma_q), \qquad p(z) = \mathcal{N}(\mu_p, \Sigma_p),$$

their KL divergence has a closed-form expression:

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left(\mathrm{tr}(\Sigma_p^{-1}\Sigma_q) + (\mu_p - \mu_q)^T \Sigma_p^{-1}(\mu_p - \mu_q) - k + \log \frac{|\Sigma_p|}{|\Sigma_q|}\right),$$

where k is the dimensionality of the latent variable z.

How:

The log density of a multivariate Gaussian is

$$\log \mathcal{N}(z|\mu, \Sigma) = -\frac{1}{2}\left(k \log(2\pi) + \log|\Sigma| + (z-\mu)^T \Sigma^{-1}(z-\mu)\right).$$

Applying this to both distributions gives

$$\log q(z) = -\frac{1}{2}\left(k \log(2\pi) + \log|\Sigma_q| + (z-\mu_q)^T \Sigma_q^{-1}(z-\mu_q)\right),$$

$$\log p(z) = -\frac{1}{2}\left(k \log(2\pi) + \log|\Sigma_p| + (z-\mu_p)^T \Sigma_p^{-1}(z-\mu_p)\right).$$

Therefore:

$$\log \frac{q(z)}{p(z)} = -\frac{1}{2}\left(\log|\Sigma_q| - \log|\Sigma_p| + (z-\mu_q)^T \Sigma_q^{-1}(z-\mu_q) - (z-\mu_p)^T \Sigma_p^{-1}(z-\mu_p)\right).$$

To take the expectation under $q(z)$, two useful identities for Gaussian expectations:

$$\mathbb{E}_q[z] = \mu_q$$

and

$$\mathbb{E}_q[(z-\mu_q)(z-\mu_q)^T] = \Sigma_q.$$

With these:

$$\mathbb{E}_q[(z-\mu_q)^T\Sigma_q^{-1}(z-\mu_q)] = \mathrm{tr}(\Sigma_q^{-1}\Sigma_q) = k.$$

Evaluating the quadratic expectation. To compute the KL divergence between two Gaussians we need to evaluate

$$\mathbb{E}_q[(z-\mu_p)^T\Sigma_p^{-1}(z-\mu_p)].$$

A useful identity for quadratic forms is

$$x^T Ax = \mathrm{tr}(A\,xx^T).$$

Using this identity we rewrite the quadratic term as

$$(z-\mu_p)^T\Sigma_p^{-1}(z-\mu_p) = \mathrm{tr}\left(\Sigma_p^{-1}(z-\mu_p)(z-\mu_p)^T\right).$$

Taking the expectation with respect to $q(z)$ gives

$$\mathbb{E}_q[(z-\mu_p)^T\Sigma_p^{-1}(z-\mu_p)] = \mathrm{tr}\left(\Sigma_p^{-1}\mathbb{E}_q[(z-\mu_p)(z-\mu_p)^T]\right).$$

We therefore need to compute

$$\mathbb{E}_q[(z-\mu_p)(z-\mu_p)^T].$$

Expanding around the mean. Write

$$z-\mu_p = (z-\mu_q) + (\mu_q-\mu_p).$$

Substituting this expression gives

$$(z-\mu_p)(z-\mu_p)^T = (z-\mu_q)(z-\mu_q)^T + (z-\mu_q)(\mu_q-\mu_p)^T + (\mu_q-\mu_p)(z-\mu_q)^T + (\mu_q-\mu_p)(\mu_q-\mu_p)^T.$$

Taking the expectation under $q(z)$ simplifies the expression because

$$\mathbb{E}_q[z-\mu_q] = 0.$$

Therefore the cross terms vanish and we obtain

$$\mathbb{E}_q[(z-\mu_p)(z-\mu_p)^T] = \mathbb{E}_q[(z-\mu_q)(z-\mu_q)^T] + (\mu_q-\mu_p)(\mu_q-\mu_p)^T.$$

The first term is simply the covariance matrix of q,

$$\mathbb{E}_q[(z-\mu_q)(z-\mu_q)^T] = \Sigma_q.$$

Thus

$$\mathbb{E}_q[(z-\mu_p)(z-\mu_p)^T] = \Sigma_q + (\mu_q-\mu_p)(\mu_q-\mu_p)^T.$$

Substituting back into the quadratic expectation. Plugging this result into the trace expression gives

$$\mathbb{E}_q[(z-\mu_p)^T\Sigma_p^{-1}(z-\mu_p)] = \mathrm{tr}\big(\Sigma_p^{-1}\left(\Sigma_q + (\mu_q-\mu_p)(\mu_q-\mu_p)^T\right)\big).$$

Using the linearity of the trace and the identity $\mathrm{tr}(A\,bb^T) = b^T Ab$ yields

$$\mathbb{E}_q[(z-\mu_p)^T\Sigma_p^{-1}(z-\mu_p)] = \mathrm{tr}(\Sigma_p^{-1}\Sigma_q) + (\mu_q-\mu_p)^T\Sigma_p^{-1}(\mu_q-\mu_p).$$

This separates the contribution of the covariance mismatch (first term) and the mean mismatch (second term).

Step 4: Combine all terms. Substituting these expectations into the KL definition gives

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left(\mathrm{tr}(\Sigma_p^{-1}\Sigma_q) + (\mu_p-\mu_q)^T\Sigma_p^{-1}(\mu_p-\mu_q) - k + \log\frac{|\Sigma_p|}{|\Sigma_q|}\right).$$

This is the closed-form KL divergence between two multivariate Gaussian distributions.

Applying this to the VAE case. For VAEs we have

$$p(z) = \mathcal{N}(0, I), \qquad q(z|x) = \mathcal{N}(\mu, \mathrm{diag}(\sigma^2)).$$

Substituting these into the Gaussian KL formula greatly simplifies the expression because

$$\Sigma_p = I$$

and

$$\Sigma_q = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_D^2).$$

The trace term becomes

$$\mathrm{tr}(\Sigma_q) = \sum_d \sigma_d^2,$$

the quadratic mean term becomes

$$\mu^T\mu = \sum_d \mu_d^2,$$

and the determinant of a diagonal covariance matrix is

$$|\Sigma_q| = \prod_d \sigma_d^2, \qquad \log|\Sigma_q| = \sum_d \log \sigma_d^2.$$

Substituting these results yields

$$D_{\mathrm{KL}}(q(z|x) \,\|\, p(z)) = \frac{1}{2}\left(\sum_d \sigma_d^2 + \sum_d \mu_d^2 - D - \sum_d \log \sigma_d^2\right).$$

Rearranging the expression per latent dimension gives the well-known VAE formula

$$D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, p(z)] = \frac{1}{2}\sum_d \left(\mu_d^2 + \sigma_d^2 - \log \sigma_d^2 - 1\right).$$

(a) Write $q(z|x,\theta)$ explicitly.

And for this case we are given the encoder outputs for one input image x:

$$\mu[x,\theta] = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}, \qquad \log \sigma^2[x,\theta] = \begin{bmatrix} -1 \\ \log 4 \end{bmatrix}.$$

This means the encoder predicts a 2-dimensional Gaussian approximate posterior. Since the covariance is diagonal, the two latent dimensions are independent under $q(z|x,\theta)$, and each one has its own mean and variance.

The vector of log-variances is

$$\log \sigma^2[x,\theta] = \begin{bmatrix} -1 \\ \log 4 \end{bmatrix}.$$

To recover the variances themselves, we exponentiate componentwise:

$$\sigma_1^2 = e^{-1}, \qquad \sigma_2^2 = e^{\log 4} = 4.$$

So the covariance matrix is

$$\mathrm{Diag}[\sigma_1^2, \sigma_2^2] = \mathrm{Diag}[e^{-1}, 4].$$

Therefore the approximate posterior is

$$\boxed{q(z|x,\theta) = \mathcal{N}\left(z;\ \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}, \mathrm{Diag}[e^{-1}, 4]\right)}$$

(b) Computing $D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z)]$

We are told that the prior is standard normal:

$$\Pr(z) = \mathcal{N}(z; 0, I).$$

For a diagonal Gaussian approximate posterior against a standard normal prior, the KL divergence has the standard formula

$$D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z)] = \frac{1}{2}\sum_{d=1}^{2}\left(\mu_d^2 + \sigma_d^2 - \log\sigma_d^2 - 1\right)$$

, which we just derived.

In our case:

$$\mu_1 = 1, \qquad \mu_2 = -0.5, \qquad \sigma_1^2 = e^{-1}, \qquad \sigma_2^2 = 4.$$

We now compute the contribution from each latent dimension separately.

Dimension $d = 1$

$$\mu_1^2 + \sigma_1^2 - \log\sigma_1^2 - 1 = 1^2 + e^{-1} - (-1) - 1 = 1 + e^{-1} + 1 - 1 = 1 + e^{-1}.$$

Dimension $d = 2$

$$\mu_2^2 + \sigma_2^2 - \log\sigma_2^2 - 1 = (-0.5)^2 + 4 - \log 4 - 1 = 3.25 - \log 4.$$

So, combining the two dimensions:

$$\boxed{D_{\mathrm{KL}}[q(z|x,\theta) \,\|\, \Pr(z)] = \frac{1}{2}\left(1 + e^{-1} + 3.25 - \log 4\right) \approx 1.616}$$

c) Using the reparameterization

$$z^* = \mu[x,\theta] + \sigma[x,\theta] \odot \epsilon, \qquad \text{where } \sigma[x,\theta] = \begin{bmatrix}\sigma_1\\ \sigma_2\end{bmatrix}, \qquad \epsilon = \begin{bmatrix}0.3\\ -1\end{bmatrix},$$

compute one sample z^*.

$\odot$ denotes elementwise multiplication, and we already know the variances, so we first compute the standard deviations:

$$\sigma_1 = \sqrt{e^{-1}} = e^{-1/2}, \qquad \sigma_2 = \sqrt{4} = 2.$$

Thus

$$\sigma[x,\theta] = \begin{bmatrix}e^{-1/2}\\ 2\end{bmatrix}.$$

We are also given

$$\epsilon = \begin{bmatrix}0.3\\ -1\end{bmatrix}.$$

Now compute the elementwise product:

$$\sigma[x,\theta] \odot \epsilon = \begin{bmatrix} e^{-1/2} \\ 2 \end{bmatrix} \odot \begin{bmatrix} 0.3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.3e^{-1/2} \\ -2 \end{bmatrix}.$$

Now add the mean vector:

$$\boxed{z^* = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix} + \begin{bmatrix} 0.3e^{-1/2} \\ -2 \end{bmatrix} = \begin{bmatrix} 1 + 0.3e^{-1/2} \\ -2.5 \end{bmatrix} \approx \begin{bmatrix} 1.182 \\ -2.5 \end{bmatrix}.}$$

Problem 5. My Nonna's Gaussian Lasagna Consider the forward diffusion process

$$\begin{aligned} z_1 &= \sqrt{1-\beta_1}\, x + \sqrt{\beta_1}\, \epsilon_1, \\ z_2 &= \sqrt{1-\beta_2}\, z_1 + \sqrt{\beta_2}\, \epsilon_2, \\ z_3 &= \sqrt{1-\beta_3}\, z_2 + \sqrt{\beta_3}\, \epsilon_3, \end{aligned}$$

where $\epsilon_1, \epsilon_2, \epsilon_3 \sim \mathcal{N}(0, I)$ are independent.

Equation 18.6 shows that

$$z_2 = \sqrt{(1-\beta_2)(1-\beta_1)}\, x + \sqrt{1-(1-\beta_2)(1-\beta_1)}\, \epsilon,$$

for some $\epsilon \sim \mathcal{N}(0, I)$.

Continue this process to show that

$$z_3 = \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, x + \sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, \epsilon',$$

where $\epsilon' \sim \mathcal{N}(0, I)$.

Theory summary: The forward diffusion process.

Diffusion models generate data by gradually adding noise to an input until it becomes pure Gaussian noise. The part of the model that performs this progressive corruption of the data is called the *forward diffusion process* (sometimes also referred to as the encoder).

We start from an observed data point

$$x = z_0.$$

At each step of the forward process we create a new latent variable z_t by slightly perturbing the previous one with Gaussian noise:

$$z_t = \sqrt{1-\beta_t}\, z_{t-1} + \sqrt{\beta_t}\, \epsilon_t,$$

where

- z_t are the **latent variables** of the diffusion process. They represent progressively noisier versions of the original data.
- β_t is a small positive number controlling how much noise is added at step t. The sequence $\{\beta_t\}$ is called the **noise schedule**.
- $\epsilon_t \sim \mathcal{N}(0, I)$ are independent samples of Gaussian noise.

At each step we therefore keep part of the previous signal ($\sqrt{1-\beta_t}\, z_{t-1}$) and add a small amount of Gaussian noise ($\sqrt{\beta_t}\, \epsilon_t$).

Progressive corruption of the data. After one step we obtain

$$z_1 = \sqrt{1-\beta_1}\, x + \sqrt{\beta_1}\, \epsilon_1.$$

After two steps,

$$z_2 = \sqrt{1-\beta_2}\, z_1 + \sqrt{\beta_2}\, \epsilon_2.$$

As the number of steps increases, more noise is injected into the system and the signal component gradually disappears. After many steps the latent variable becomes approximately pure Gaussian noise:

$$z_T \sim \mathcal{N}(0, I).$$

Probabilistic interpretation. Each diffusion step defines a conditional distribution

$$q(z_t|z_{t-1}) = \mathcal{N}\big(\sqrt{1-\beta_t}\, z_{t-1}, \beta_t I\big).$$

This means that the entire forward process defines a Markov chain

$$q(z_{1:T}|x) = \prod_{t=1}^{T} q(z_t|z_{t-1}).$$

Closed-form expression. A key property of this process is that because each step is Gaussian, we can directly write the distribution of z_t conditioned on the original data x. After t steps,

$$z_t = \sqrt{\bar{\alpha}_t}\, x + \sqrt{1-\bar{\alpha}_t}\, \epsilon,$$

where

$$\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i), \qquad \epsilon \sim \mathcal{N}(0, I).$$

This result shows that after multiple diffusion steps the latent variable is simply a weighted combination of the original signal x and Gaussian noise.

This property is key in diffusion models because it allows us to sample z_t directly from x without simulating all intermediate steps, which greatly simplifies both training and analysis of the model.

What the exercice specifically asks: Let's prove one of the steps of this proof: we start from the recursion relation for the third diffusion step:

$$z_3 = \sqrt{1-\beta_3}\, z_2 + \sqrt{\beta_3}\, \epsilon_3.$$

We are also given that

$$z_2 = \sqrt{(1-\beta_2)(1-\beta_1)}\, x + \sqrt{1-(1-\beta_2)(1-\beta_1)}\, \epsilon,$$

where $\epsilon \sim \mathcal{N}(0, I)$.

And we need to show that:

$$z_3 = \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, x + \sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, \epsilon',$$

where $\epsilon' \sim \mathcal{N}(0, I)$.

Replacing z_2 in the expression for $z_3 = \sqrt{1-\beta_3} z_2 + \sqrt{\beta_3}\epsilon_3$, we get

$$z_3 = \sqrt{1-\beta_3}\left[\sqrt{(1-\beta_2)(1-\beta_1)}\, x + \sqrt{1-(1-\beta_2)(1-\beta_1)}\, \epsilon\right] + \sqrt{\beta_3}\, \epsilon_3.$$

Now distribute the factor $\sqrt{1-\beta_3}$:

$$z_3 = \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, x + \sqrt{(1-\beta_3)\Big(1-(1-\beta_2)(1-\beta_1)\Big)}\, \epsilon + \sqrt{\beta_3}\, \epsilon_3.$$

The first term is the deterministic part multiplying x. The remaining two terms are both Gaussian noise terms:

$$\sqrt{(1-\beta_3)\Big(1-(1-\beta_2)(1-\beta_1)\Big)}\, \epsilon \qquad \text{and} \qquad \sqrt{\beta_3}\, \epsilon_3.$$

Since $\epsilon \sim \mathcal{N}(0, I)$ and $\epsilon_3 \sim \mathcal{N}(0, I)$ are independent, their weighted sum is again Gaussian with mean zero.

This is because Gaussian distributions are closed under linear transformations (these kind of basic properties for the main distributions are quite useful, will try to get them all together, but it might not be complete), and if you compute the mean and variance for this it gives the ones above.

So we now compute the covariance of this combined Gaussian noise term: Let

$$A = \sqrt{(1-\beta_3)\Big(1-(1-\beta_2)(1-\beta_1)\Big)}, \qquad B = \sqrt{\beta_3}.$$

Then the noise part can be written as

$$A\, \epsilon + B\, \epsilon_3.$$

Because ϵ and ϵ_3 are independent standard Gaussians, we have

$$A\, \epsilon + B\, \epsilon_3 \sim \mathcal{N}\big(0, (A^2 + B^2) I\big).$$

So to get the variance we compute:

$$A^2 + B^2 = (1-\beta_3)\Big(1-(1-\beta_2)(1-\beta_1)\Big) + \beta_3 = 1-(1-\beta_3)(1-\beta_2)(1-\beta_1).$$

Therefore the total noise term has distribution

$$\mathcal{N}\left(0, [1-(1-\beta_3)(1-\beta_2)(1-\beta_1)]\, I\right).$$

This means we can write it as

$$\sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, \epsilon',$$

where

$$\epsilon' \sim \mathcal{N}(0, I).$$

Substituting this back into the expression for z_3, we obtain

$$\boxed{z_3 = \sqrt{(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, x + \sqrt{1-(1-\beta_3)(1-\beta_2)(1-\beta_1)}\, \epsilon', \qquad \epsilon' \sim \mathcal{N}(0, I)}$$

Problem 6. Buy \$DIRAC coin Consider a 1D diffusion model where the data distribution consists of two point masses describing draws of a fair coin:

$$\Pr(x) = \frac{1}{2}\,\delta[x-1] + \frac{1}{2}\,\delta[x+1].$$

We are interested in building a diffusion model that can describe this in terms of latents $z_1, \ldots, z_T$, with $\Pr(z_T) \equiv \mathcal{N}(0,1)$ and some model family for $\Pr(z_{t-1} \mid z_t)$.

Define the forward diffusion transitions by

$$q(z_t \mid z_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, z_{t-1},\, \beta_t\Big),$$

where $\beta_t \in [0, 1]$ is the noise schedule. This implies the diffusion kernel

$$q(z_t \mid x) = \mathcal{N}(\sqrt{\alpha_t}\, x,\, 1-\alpha_t),$$

where

$$\alpha_t = \prod_{s=1}^{t}(1-\beta_s).$$

The forward process defines a joint distribution over $(x, z_{1:T})$: first draw $x \sim \Pr(x)$, then run the Markov chain (note that we choose to define $q(x) = Pr(x)$.

(a) Write down the marginal distribution $q(z_t)$ by marginalizing over x. Describe qualitatively how this distribution changes as t goes from 1 to T.

We are given the conditional diffusion kernel

$$q(z_t \mid x) = \mathcal{N}(\sqrt{\alpha_t}\, x,\, 1-\alpha_t),$$

and the data distribution

$$\Pr(x) = \frac{1}{2}\delta[x-1] + \frac{1}{2}\delta[x+1].$$

To obtain the marginal $q(z_t)$, we average over the two possible values of x:

$$q(z_t) = \sum_{x \in \{-1,+1\}} q(z_t \mid x)\Pr(x).$$

Since each value of x has probability 1/2 of being 1 or -1, this becomes

$$q(z_t) = \frac{1}{2}\mathcal{N}(z_t; \sqrt{\alpha_t},\, 1-\alpha_t) + \frac{1}{2}\mathcal{N}(z_t; -\sqrt{\alpha_t},\, 1-\alpha_t).$$

So the marginal distribution is a *mixture of two Gaussians*, centered at $\pm\sqrt{\alpha_t}$, both with variance $1-\alpha_t$.

Qualitative behavior as t increases. At early times, α_t is close to 1, so:

$$\sqrt{\alpha_t} \approx 1, \qquad 1 - \alpha_t \approx 0.$$

This means $q(z_t)$ looks like two narrow peaks near $z_t = \pm 1$, which closely resemble the original two-point distribution.

As t increases, noise is added:

- the means $\pm\sqrt{\alpha_t}$ move closer to 0: Recall that

  $$\alpha_t = \prod_{s=1}^{t} (1 - \beta_s).$$

  Since the noise schedule satisfies $0 < \beta_s < 1$, each factor $0 < (1 - \beta_s) < 1$. Multiplying many numbers strictly smaller than 1 makes the product decrease with t, so

  $$\alpha_t \to 0 \quad \text{as } t \text{ increases.}$$

- the variance $1 - \alpha_t$ increases to 1 as α_t decreases to 0.

Eventually, as $t \to T$, we typically have $\alpha_t \to 0$, so both components collapse to the same Gaussian:

$$q(z_T) \approx \mathcal{N}(0, 1).$$

Thus, the forward process gradually transforms the original discrete two-point distribution into an approximately standard normal distribution.

(b) Using Bayes' rule and the diffusion kernel, show that

$$q(x = 1 \mid z_t) = \operatorname{sig}\left[\frac{2\sqrt{\alpha_t}\, z_t}{1 - \alpha_t}\right],$$

and similarly for $q(x = -1 \mid z_t)$, where

$$\operatorname{sig}(u) = \frac{1}{1 + e^{-u}}$$

is the logistic sigmoid.

We use Bayes' rule:

$$q(x = 1 \mid z_t) = \frac{q(z_t \mid x = 1)\,\Pr(x = 1)}{q(z_t \mid x = 1)\,\Pr(x = 1) + q(z_t \mid x = -1)\,\Pr(x = -1)}.$$

Because $\Pr(x = 1) = \Pr(x = -1) = 1/2$, the factors of $1/2$ cancel, giving

$$q(x = 1 \mid z_t) = \frac{q(z_t \mid x = 1)}{q(z_t \mid x = 1) + q(z_t \mid x = -1)}.$$

Now substitute the two Gaussian densities:

$$q(z_t \mid x = 1) = \frac{1}{\sqrt{2\pi(1 - \alpha_t)}} \exp\left(-\frac{(z_t - \sqrt{\alpha_t})^2}{2(1 - \alpha_t)}\right),$$

$$q(z_t \mid x=-1) = \frac{1}{\sqrt{2\pi(1-\alpha_t)}} \exp\left(-\frac{(z_t+\sqrt{\alpha_t})^2}{2(1-\alpha_t)}\right).$$

The normalization constants cancel in the ratio, so

$$q(x=1 \mid z_t) = \frac{\exp\left(-\frac{(z_t-\sqrt{\alpha_t})^2}{2(1-\alpha_t)}\right)}{\exp\left(-\frac{(z_t-\sqrt{\alpha_t})^2}{2(1-\alpha_t)}\right) + \exp\left(-\frac{(z_t+\sqrt{\alpha_t})^2}{2(1-\alpha_t)}\right)}.$$

Factor out the first exponential from numerator and denominator:

$$q(x=1 \mid z_t) = \frac{1}{1+\exp\left[-\frac{(z_t+\sqrt{\alpha_t})^2-(z_t-\sqrt{\alpha_t})^2}{2(1-\alpha_t)}\right]}.$$

Now simplify the difference in squares:

$$(z_t+\sqrt{\alpha_t})^2 - (z_t-\sqrt{\alpha_t})^2 = 4\sqrt{\alpha_t}\, z_t.$$

Therefore

$$q(x=1 \mid z_t) = \frac{1}{1+\exp\left[-\frac{2\sqrt{\alpha_t}\, z_t}{1-\alpha_t}\right]} = \mathrm{sig}\left(\frac{2\sqrt{\alpha_t}\, z_t}{1-\alpha_t}\right).$$

Similarly,

$$q(x=-1 \mid z_t) = 1 - q(x=1 \mid z_t) = \mathrm{sig}\left(-\frac{2\sqrt{\alpha_t}\, z_t}{1-\alpha_t}\right).$$

(c) Using parts (a) and (b), show that the true reverse distribution $q(z_{t-1} \mid z_t)$ can be written as a weighted mixture of two Gaussians, corresponding to the two possible values $x = \pm 1$.

We want to express the reverse conditional $q(z_{t-1} \mid z_t)$. Since the only possible values of x are ± 1, we can condition on x and sum over these two cases (law of total probability, we marginalize over x (discrete case)), and then we use the product rule to factorize the joint distribution:

$$q(z_{t-1} \mid z_t) = \sum_{x\in\{-1,+1\}} q(z_{t-1}, x \mid z_t) = \sum_{x\in\{-1,+1\}} q(z_{t-1} \mid z_t, x)\, q(x \mid z_t).$$

So the reverse distribution is a weighted mixture of the two conditionals $q(z_{t-1} \mid z_t, x=1)$ and $q(z_{t-1} \mid z_t, x=-1)$, with weights $q(x=1 \mid z_t)$ and $q(x=-1 \mid z_t)$.

Now, for fixed x, the pair (z_{t-1}, z_t) is jointly Gaussian, so $q(z_{t-1} \mid z_t, x)$ is Gaussian.

First, from the diffusion kernel at step $t-1$,

$$q(z_{t-1} \mid x) = \mathcal{N}(\sqrt{\alpha_{t-1}}\, x,\ 1-\alpha_{t-1}).$$

Next, the forward step from z_{t-1} to z_t is

$$q(z_t \mid z_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\, z_{t-1}, \beta_t\right).$$

Computing $q(z_{t-1} \mid z_t, x)$ using the standard Gaussian conditioning formula

We now derive the conditional distribution

$$q(z_{t-1} \mid z_t, x).$$

Using Bayes' rule, the desired conditional satisfies

$$q(z_{t-1} \mid z_t, x) \propto q(z_t \mid z_{t-1})\, q(z_{t-1} \mid x).$$

Writing both terms as functions of z_{t-1}

We now view both densities as functions of z_{t-1}.

The first term can be written as

$$q(z_t \mid z_{t-1}) \propto \exp\left(-\frac{1}{2\beta_t} \left\| z_t - \sqrt{1-\beta_t}\, z_{t-1} \right\|^2\right),$$

and the second term as

$$q(z_{t-1} \mid x) \propto \exp\left(-\frac{1}{2(1-\alpha_{t-1})} \left\| z_{t-1} - \sqrt{\alpha_{t-1}}\, x \right\|^2\right).$$

Multiplying these two expressions corresponds to adding the quadratic terms in the exponent.

Combining the quadratic terms

Expanding both squares and collecting the terms involving z_{t-1}, we obtain a quadratic expression of the form

$$-\frac{1}{2}\left[A\, z_{t-1}^2 - 2B\, z_{t-1}\right],$$

where

$$A = \frac{1-\beta_t}{\beta_t} + \frac{1}{1-\alpha_{t-1}}, \qquad B = \frac{\sqrt{1-\beta_t}}{\beta_t} z_t + \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_{t-1}} x.$$

Completing the square

A quadratic of this form corresponds to a Gaussian distribution. Completing the square gives

$$q(z_{t-1} \mid z_t, x) = \mathcal{N}\left(z_{t-1}; \frac{B}{A}, \frac{1}{A}\right).$$

So the mean and variance are

$$\mu_x(z_t) = \frac{B}{A}, \qquad \sigma^2_{\text{comp}} = \frac{1}{A}.$$

Simplifying the expression

Using the identity

$$\alpha_t = \alpha_{t-1}(1-\beta_t),$$

and simplifying the algebra, we obtain

$$q(z_{t-1} \mid z_t, x) = \mathcal{N}\big(z_{t-1};\, \mu_x(z_t),\, \sigma^2_{\mathrm{comp}}\big)\,,$$

with

$$\mu_x(z_t) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t}\, z_t + \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}\, x,$$

and

$$\sigma^2_{\mathrm{comp}} = \frac{(1-\alpha_{t-1})\beta_t}{1-\alpha_t}.$$

The mean is a weighted combination of the noisy observation z_t and the original signal x, while the variance reflects the uncertainty remaining after combining both sources of information.

Therefore

$$q(z_{t-1} \mid z_t) = q(x=1 \mid z_t)\,\mathcal{N}\big(z_{t-1}; \mu_+(z_t), \sigma^2_{\mathrm{comp}}\big) + q(x=-1 \mid z_t)\,\mathcal{N}\big(z_{t-1}; \mu_-(z_t), \sigma^2_{\mathrm{comp}}\big)\,,$$

where

$$\mu_\pm(z_t) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t}\, z_t \pm \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}.$$

Hence the true reverse distribution is indeed a *weighted mixture of two Gaussians*, one corresponding to $x=+1$ and one to $x=-1$.

Faster way to do this: formula for conditioning in a joint Gaussian

Given

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right).$$

Then the conditional distribution of x_1 given x_2 is

$$p(x_1 \mid x_2) = \mathcal{N}\big(x_1; \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2-\mu_2),\ \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\big)\,.$$

Applying this to our case

In our setting, the pair (z_{t-1}, z_t) is jointly Gaussian when we condition on x, because both

$$q(z_{t-1} \mid x) \quad \text{and} \quad q(z_t \mid z_{t-1})$$

are Gaussian and the model is linear.

Therefore we can identify

$$x_1 = z_{t-1}, \qquad x_2 = z_t,$$

and compute the mean and covariance terms from the forward process.

Plugging these into the Gaussian conditioning formula gives the result

$$q(z_{t-1} \mid z_t, x) = \mathcal{N}\big(z_{t-1};\, \mu_x(z_t),\, \sigma^2_{\text{comp}}\big)\,,$$

with

$$\mu_x(z_t) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t}\, z_t + \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}\, x,$$

and

$$\sigma^2_{\text{comp}} = \frac{(1-\alpha_{t-1})\beta_t}{1-\alpha_t}.$$

(d) Let $\Delta\mu = |\mu_+ - \mu_-|$ denote the distance between the two mixture component means in part (c), and let σ_{comp} denote their shared standard deviation. Show that the separation measured in standard deviations is

$$\frac{\Delta\mu}{\sigma_{\text{comp}}} = \frac{2\sqrt{\alpha_{t-1}\beta_t}}{\sqrt{(1-\alpha_t)(1-\alpha_{t-1})}},$$

and that this vanishes as $\beta_t \to 0$ with α_{t-1} held fixed.

Computing the separation in units of standard deviation

From part (c), the two Gaussian components have means

$$\mu_+(z_t) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t}\, z_t + \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t},$$

$$\mu_-(z_t) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t}\, z_t - \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}.$$

Distance between the means

The terms involving z_t cancel when we subtract, so

$$\Delta\mu = |\mu_+ - \mu_-| = \left|\frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t} - \left(-\frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}\right)\right|.$$

$$= \left|\frac{2\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}\right| = \frac{2\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}.$$

Standard deviation

Their shared variance is

$$\sigma^2_{\text{comp}} = \frac{(1-\alpha_{t-1})\beta_t}{1-\alpha_t},$$

so the standard deviation is

$$\sigma_{\text{comp}} = \sqrt{\frac{(1-\alpha_{t-1})\beta_t}{1-\alpha_t}}.$$

Forming the ratio

Separation in units of standard deviation:

$$\frac{\Delta\mu}{\sigma_{\text{comp}}} = \frac{\dfrac{2\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t}}{\sqrt{\dfrac{(1-\alpha_{t-1})\beta_t}{1-\alpha_t}}} = \frac{2\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t} \cdot \sqrt{\frac{1-\alpha_t}{(1-\alpha_{t-1})\beta_t}} = \frac{2\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t} \cdot \frac{\sqrt{1-\alpha_t}}{\sqrt{(1-\alpha_{t-1})\beta_t}}.$$

Therefore:

$$\boxed{\frac{\Delta\mu}{\sigma_{\text{comp}}} = 2 \cdot \frac{\sqrt{\alpha_{t-1}}}{\sqrt{1-\alpha_{t-1}}} \cdot \sqrt{\beta_t} \cdot \frac{1}{\sqrt{1-\alpha_t}} = \frac{2\sqrt{\alpha_{t-1}\beta_t}}{\sqrt{(1-\alpha_t)(1-\alpha_{t-1})}}}.$$

Why this goes to zero as $\beta_t \to 0$. By definition,

$$\alpha_t = \alpha_{t-1}(1-\beta_t).$$

Given that:

$$\alpha_t = \prod_{s=1}^{t}(1-\beta_s) = \left(\prod_{s=1}^{t-1}(1-\beta_s)\right)(1-\beta_t) = \alpha_{t-1}(1-\beta_t).$$

So α_{t-1} depends only on the previous steps $(\beta_1, \dots, \beta_{t-1})$, which are already fixed. When we take the limit $\beta_t \to 0$, we are only changing the noise added at step t, not the earlier steps. Therefore α_{t-1} remains constant.

Moreover,

$$\alpha_t \to \alpha_{t-1} \quad \text{as} \quad \beta_t \to 0,$$

so the denominator stays finite.

In the numerator we have $\sqrt{\beta_t}$, while the denominator remains finite. Therefore

$$\frac{\Delta\mu}{\sigma_{\text{comp}}} \to 0.$$

This means that the distance between the two Gaussian components, measured in units of their standard deviation, goes to zero. In other words, the two Gaussians become indistinguishable and effectively merge into a single Gaussian.

Why this justifies the Gaussian reverse model

From part (c), the true reverse distribution is a mixture of two Gaussians. However, when β_t is very small, these two components overlap almost perfectly, so the mixture is well approximated by a single Gaussian.

This explains why, for sufficiently small β_t, we can model the reverse step as

$$\Pr(z_{t-1} \mid z_t, \phi_t) = \mathcal{N}\big(z_{t-1};\, f_t(z_t, \phi_t),\, \sigma_t^2 I\big)\,,$$

where $f_t(z_t, \phi_t)$ is a learned mean function.

In other words, the complex mixture distribution collapses to a single Gaussian in the small-step limit, which is why Gaussian reverse models are accurate in diffusion models.